

The economic burden of non-influenza-related viral respiratory tract infection in the United States.



BACKGROUND: Viral respiratory tract infection (VRTI) is the most common illness in humans. Despite the high incidence, the economic impact of non-influenza-related VRTI has not been rigorously explored. Our objectives were to obtain an updated incidence of non-influenza-related VRTI in the United States and to quantify the health care resource use (direct costs) and productivity losses (indirect costs) associated with these infections. **METHODS:** A nationwide telephone survey of US households (N = 4051) was conducted between November 3, 2000, and February 12, 2001 to obtain a representative estimate of the self-reported incidence of non-influenza-related VRTI and related treatment patterns. Direct treatment costs measured included outpatient clinician encounters, use of over-the-counter and prescription drugs, and associated infectious complications of non-influenza-related VRTI. Absenteeism estimates for infected individuals and parents of infected children were extrapolated from National Health Interview Survey data. **RESULTS:** Of survey respondents, 72% reported a non-influenza-related VRTI within the past year. Respondents who experienced a self-reported non-influenza-related VRTI averaged 2.5 episodes annually. When these rates are extrapolated to the entire US population, approximately 500 million non-influenza-related VRTI episodes occur per year. Similarly, if the treatment patterns reported by the respondents are extended to the population, the total economic impact of non-influenza-related VRTI approaches \$40 billion annually (direct costs, \$17 billion per year; and indirect costs, \$22.5 billion per year). **CONCLUSIONS:** Largely because of the high attack rate, non-influenza-related VRTI imposes a greater economic burden than many other clinical conditions. The pending availability of effective antiviral therapies warrants increased attention be paid to this common and expensive illness.